Second-order asymptotics for edge-disjoint clique packings

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Abstract

For a fixed integer $r \geq 3$, let $\phi_r(n,k)$ be the minimum, over all graphs with n vertices and $t_{r-1}(n)+k$ edges, of the maximum size of an edge-disjoint K_r-packing, where $t_{r-1}(n)$ is the Turán number of K_r. Győri proved that $\phi_r(n,k) \geq k - O_r(k^2/n^2)$ when $k = o(n^2)$. We determine the optimal coefficient of the error term whenever $k/n \to \infty$ and $k/n^2 \to 0$. Namely, $\phi_r(n,k) = k-(A_r+o(1))k^2/n^2$, where $A_r = (r-1)^2(\sum_{j=1}^{r} j^{-2} - 3/2 + 1/r - 1/(2r(r-1)))$. The same expansion holds for fractional packings. The integer proof reduces the rounding problem to a graph with $O_r(k/n)$ vertices and extends every packing of this smaller graph without loss. For feasible positive sequences $k = k(n)$, we also show that $\phi_r(n,k)/k \to 1$ holds precisely when $k = o(n^2)$.

1 Introduction

Turán's theorem guarantees a copy of K_r as soon as the number of edges exceeds $t_{r-1}(n)$. A natural strengthening asks how many edge-disjoint copies are forced by a prescribed excess. Write $\nu_r(G)$ for the maximum size of an edge-disjoint K_r-packing in G, and set

$$\phi_r(n,k) = \min\{\nu_r(G) : v(G) = n,\ e(G) = t_{r-1}(n)+k\}.$$

Here $t_q(n)$ is the number of edges in the balanced complete q-partite graph $T_q(n)$. An integer k is feasible if $0 \leq k \leq \binom{n}{2} - t_{r-1}(n)$.

The packing problem is closely related to decomposing a graph into cliques and single edges. Erdős, Goodman, and Pósa [6] proved that every n-vertex graph has a decomposition into at most $t_2(n)$ triangles and single edges. A decomposition with m triangles has $e(G) - 2m$ members, so their theorem gives $\phi_3(n,k) \geq k/2$. Bollobás [4] proved the corresponding covering result for copies of K_r and single edges. For decompositions, Győri and Tuza [12] proved that, when $r \geq 4$, the sum of the orders of the members can be bounded by $2t_{r-1}(n)$. Since a decomposition with m copies of K_r has total order $2e(G) - r(r-2)m$, their result gives $\phi_r(n,k) \geq 2k/[r(r-2)]$.

Allowing cliques of arbitrary orders leads to weighted decomposition problems. Balogh, He, Krueger, Nguyen, and Wigal [1, Theorem 1.4] proved that every n-vertex graph admits an edge decomposition into cliques with total cost at most $n^2/4 + o(n^2)$, where a clique C has cost $|V(C)|-1$. This establishes an asymptotic form of a conjecture of Erdős. In a recent preprint, Zhang [19, Theorem 1.4] proved the exact bound $t_2(n)$ for all sufficiently large n.

Near the Turán threshold, the packing number is much closer to the excess. Early work on the triangle problem includes Erdős [5] and Sauer [17]. Erdős asked how the number of edge-disjoint

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triangles depends on the excess above $t_2(n)$, including when this excess itself can be attained. Győri proved the following bound for triangles in [8] and for larger cliques in [9]. For every fixed $r \geq 3$,

$$\phi_r(n,k) \geq k - O_r(k^2/n^2) \qquad \text{when } k = o(n^2). \tag{1}$$

The bound is recorded as Theorem 8 in [10, p. 361]; a uniform finite formulation appears in [15, Theorem 2.2]. Thus the leading term is k throughout the subquadratic range. Győri also determined ranges of linear excess for which the exact equality $\phi_r(n,k) = k$ holds; see [8, 9] and the correction to the triangle case in [10, Remark 2].

For arbitrary excess, progress on triangle packings again came through decompositions. Král', Lidický, Martins, and Pehova [14] proved that every n-vertex graph can be decomposed into edges and triangles whose total order is at most $n^2/2 + o(n^2)$. Blumenthal, Lidický, Pehova, Pfender, Pikhurko, and Volec [3] subsequently determined the sharp bound and characterized the extremal graphs for all sufficiently large n. In particular, their result bounds the total order by $n^2/2 + 1$. In terms of packings, this yields $\phi_3(n,k) \geq 2k/3 - O(1)$. Balogh and Wigal [2] proved $\phi_r(n,k) \geq (2/r - o(1))k$ for every fixed $r \geq 3$, resolving Győri's asymptotic conjecture for general cliques.

Fractional packings provide the link between these global bounds and their integer counterparts. A fractional K_r-packing assigns a nonnegative weight to every copy of K_r, with total weight at most one on every edge. Its maximum total weight is denoted by $\nu_r^*(G)$. We define $\phi_r^*(n,k)$ by replacing ν_r with ν_r^* in the definition of $\phi_r(n,k)$. Haxell and Rödl [13] proved that $\nu_r^*(G) - \nu_r(G) = o(n^2)$ uniformly over all n-vertex graphs; Yuster [18] extended this to packings by families of graphs. The main inequality of Balogh and Wigal [2] is the finite fractional bound

$$\nu_r^*(G) \geq \frac{2}{r}\left(e(G) - \frac{r-2}{2(r-1)}n^2\right).$$

Their proof uses fractional symmetrization to reduce the inequality to complete multipartite graphs. Together with Haxell–Rödl rounding and Győri's small-excess estimate, this inequality gives their asymptotic integer bound for all feasible positive excesses.

These results leave a finer question near the Turán threshold. The estimate (1) bounds the loss from k by a term of order k^2/n^2, but does not specify its optimal coefficient. In this paper we determine that coefficient for every fixed clique size whenever $k/n \to \infty$ and $k/n^2 \to 0$. The first condition makes the second-order term tend to infinity, separating it from the bounded corrections at the linear scale. For $H_r^{(2)} = \sum_{j=1}^{r} j^{-2}$, put

$$A_r = (r-1)^2\left(H_r^{(2)} - \frac{3}{2} + \frac{1}{r} - \frac{1}{2r(r-1)}\right). \tag{2}$$

Theorem 1.1. *Fix an integer $r \geq 3$. For every feasible sequence $k = k(n)$ such that $k/n \to \infty$ and $k/n^2 \to 0$,*

$$\phi_r(n,k) = k - (A_r + o(1))\frac{k^2}{n^2}, \qquad \phi_r^*(n,k) = k - (A_r + o(1))\frac{k^2}{n^2}.$$

In particular, $A_3 = 4/9$, $A_4 = 19/16$, and $A_5 = 499/225$. The formula also gives $A_r/r^2 \to \pi^2/6 - 3/2$ as $r \to \infty$; the asymptotic statements in Theorem 1.1 are for fixed r. The next theorem gives a uniform fractional lower bound, including excesses for which the second-order term is bounded. This uniformity is also needed after the reductions in the integer proof.

Theorem 1.2. *For every fixed $r \geq 3$ there are constants $c_r, B_r > 0$ such that every graph on $n \geq 1$ vertices with $e(G) = t_{r-1}(n) + k$ and $0 \leq k \leq c_r n^2$ satisfies*

$$\nu_r^*(G) \geq k - A_r\frac{k^2}{n^2} - B_r\frac{k^{5/2}}{n^3} - B_r.$$

Balogh and Wigal [2, Section 4] asked for the range of k in which $\phi_r(n,k) = (1-o(1))k$. The next theorem determines this range for both integer and fractional packings.

Theorem 1.3. *Fix $r \geq 3$, and let $k = k(n)$ be a feasible positive integer sequence. Then each of the statements*

$$\frac{\phi_r(n,k)}{k} \longrightarrow 1, \qquad \frac{\phi_r^*(n,k)}{k} \longrightarrow 1$$

is equivalent to $k = o(n^2)$.

The sufficiency in Theorem 1.3 follows from Győri's bound. For the converse, the construction supplies a deficit at every prescribed edge count in the positive-density range. The constructions are joins of a small clique and a complete multipartite graph. Such joins already occur in Sauer's triangle construction, reported by Erdős [5, p. 98], and in [10, Example 1]. The unbalanced part sizes and the covering weights used here produce the coefficient (2).

The presence of larger cliques matters even for the first-order problem. Győri and Keszegh [11] proved that every K_4-free graph with $t_2(n) + k$ edges contains at least k edge-disjoint triangles. Thus no loss is needed under this restriction, whereas Theorem 1.1 gives a loss with coefficient $4/9$ for general graphs in the subquadratic range considered here. The small clique in the constructions above supplies the obstruction excluded by the K_4-free hypothesis.

The main obstacle in passing from fractional to integer packings is the scale of the error. Applying Haxell–Rödl rounding directly to G gives an error $o(n^2)$, which need not be smaller than k^2/n^2. Indeed, Balogh, He, Krueger, Nguyen, and Wigal [1, Theorem 5.8] constructed, for every fixed $r \geq 3$, n-vertex graphs with $\nu_r^*(G) - \nu_r(G) \geq n^{2-o(1)}$. Thus a bound of smaller polynomial order is unavailable for arbitrary graphs. The lower-bound arguments instead isolate a set of $O_r(k/n)$ exceptional vertices after removing edges or packed cliques whose contribution has already been accounted for. A clique-counting argument based on Moon and Moser [16] provides the initial structure. Related quadratic bounds on the number of edges that must be deleted to make a near-extremal clique-free graph multipartite go back to Győri [9]; see the discussion following Corollary 2 in Füredi [7]. We then control the missing cross-edges and the quadratic cost of the imbalance of the external parts at the scale of the square of the exceptional-set size. Optimizing the fractional covering constraints yields the coefficient A_r.

For the integer bound, we first absorb internal edges and then complete selected missing edges. Adding an edge increases the maximum packing number by at most one, so completion cannot decrease the deficit measured from the updated edge excess. The resulting graph has an arbitrary neighborhood in at most one external part at each exceptional vertex. Its fractional packing number splits exactly into an explicit baseline and the fractional packing number of a graph with $O_r(k/n)$ vertices. Every integer packing of this smaller graph extends with the same baseline contribution. Applying Haxell–Rödl rounding there therefore gives an error $o(k^2/n^2)$, even when k/n tends to infinity arbitrarily slowly.

Section 2 records the preliminary facts, gives the extremal constructions, and proves Theorem 1.3. Section 3 establishes the fractional lower bound. Section 4 completes the integer proof by reducing the rounding problem to a smaller graph. Section 5 discusses the remaining questions.

2 Preliminaries and constructions

We collect the duality and structural facts used in both lower-bound arguments. The constructions then give the upper bound in Theorem 1.1 and the converse in Theorem 1.3.

Throughout the paper, graphs are finite and simple and $r \geq 3$ is fixed. We put $q = r-1$, $\rho = (q-1)/(2q)$, and $g_q(n) = \rho n^2 - t_q(n)$. If $n \equiv a \pmod q$, where $0 \leq a < q$, then $g_q(n) =$

$a(q-a)/(2q)$; in particular $0 \le g_q(n) \le q/8$. Implicit constants may depend on r. For sequences we write $n \ll k \ll n^2$ to mean $k/n \to \infty$ and $k/n^2 \to 0$.

A fractional K_r-cover assigns weights $w_e \ge 0$ to the edges so that every K_r has total edge weight at least one. Linear programming duality identifies its minimum cost with $\nu_r^*(G)$. Weights larger than one can be replaced by one. We use two consequences of duality.

Lemma 2.1. *Let w be a minimum fractional K_r-cover of G with weights in $[0,1]$. Delete every edge of weight one to obtain G_0. Then the restricted cover is optimal in G_0, and*

$$e(G_0) - \nu_r^*(G_0) = e(G) - \nu_r^*(G).$$

Moreover, the number of positive-weight edges in G_0 is at most $\binom{r}{2}\nu_r^*(G_0)$.

Proof. If a edges are deleted, restriction gives a cover of cost $\nu_r^*(G) - a$. Conversely, a cover of G_0 extends to a cover of G by giving weight one to each deleted edge. These two inequalities prove the asserted equality and optimality. Complementary slackness with an optimal fractional packing shows that every positive-weight edge has packing load one. Summing these loads bounds the number of such edges by $\binom{r}{2}\nu_r^*(G_0)$. □

The next lemma extracts the small exceptional set from the clique-ratio inequality of Moon and Moser.

Lemma 2.2. *Let $q \ge 2$, and let F be a K_{q+1}-free graph on n vertices with $e(F) > t_{q-1}(n)$, where $t_1(n) = 0$. Put $\Delta = (q-1)n^2/(2q) - e(F)$. There is a q-clique $Q = \{v_1, \dots, v_q\}$ such that the sets*

$$A_i = \bigcap_{j \ne i} N_F(v_j) \qquad (1 \le i \le q)$$

are disjoint independent sets in F and satisfy

$$n - \sum_{i=1}^{q} |A_i| \le \frac{2q(q-1)\Delta}{n}.$$

Proof. Let c_j denote the number of j-cliques in F, and let $d(C)$ be the common-neighborhood size of a clique C. Turán's theorem gives $c_q > 0$, so all the clique counts below are positive. The Moon–Moser inequality [16, p. 285, equation (10)] gives

$$\frac{(j+1)c_{j+1}}{jc_j} \ge \frac{jc_j}{(j-1)c_{j-1}} - \frac{n}{j(j-1)} \qquad (2 \le j < q).$$

Iterating from $c_1 = n$ and $c_2 = e(F)$, we obtain

$$\frac{qc_q}{(q-1)c_{q-1}} \ge \frac{2e(F)}{n} - \frac{q-2}{q-1}n.$$

For $q = 2$ this is an equality. Double counting and Cauchy–Schwarz now give

$$\frac{1}{c_q} \sum_{Q \cong K_q} \sum_{\substack{C \subset Q \\ |C| = q-1}} d(C) = \frac{\sum_{C \cong K_{q-1}} d(C)^2}{c_q} \ge \frac{q^2 c_q}{c_{q-1}} \ge n - \frac{2q(q-1)\Delta}{n}.$$

Choose Q attaining at least this average. Two of the sets A_i cannot meet, since a common vertex would extend Q to a K_{q+1}. An edge inside A_i, together with $Q \setminus \{v_i\}$, would also give a K_{q+1}. Thus the sets are disjoint and independent, and their total size has the claimed lower bound. □

We use the following uniform form of the Haxell–Rödl theorem; see also [18, Theorem 1.1].

Theorem 2.3 (Haxell–Rödl [13])**.** *For every fixed r and every $\varepsilon > 0$ there exists N_0 such that every graph J on $N \geq N_0$ vertices satisfies*

$$0 \leq \nu_r^*(J) - \nu_r(J) \leq \varepsilon N^2.$$

We turn to the constructions, beginning with a deficit at every prescribed edge count. The final adjustment adds edges of covering weight one, preserving the difference between the edge excess and the cover cost.

Let $G_t = K_t \vee T_q(n-t)$, where the join adds all edges between the two graphs. Give each edge inside K_t weight $2/(rq)$, each edge from K_t to the external graph weight $1/q$, and each external edge weight zero. A copy of K_r with j vertices in K_t has weight

$$\binom{j}{2}\frac{2}{rq} + \frac{j(r-j)}{q} = \frac{j(r+1-j)}{r} = 1 + \frac{(j-1)(r-j)}{r} \geq 1.$$

Thus the cover cost is $P_t = t(n-t)/q + t(t-1)/(rq)$. Writing $k_t = e(G_t) - t_q(n)$, direct counting gives

$$\begin{aligned} k_t &= \frac{nt}{q} - \frac{t^2}{2q} - \frac{t}{2} + g_q(n) - g_q(n-t), \\ k_t - P_t &= \frac{r-2}{2rq}\, t(t-r-1) + g_q(n) - g_q(n-t). \end{aligned} \tag{3}$$

These graphs can be nested by moving a vertex from a largest external part to the clique. At step t this adds $\lfloor (n-t-1)/q \rfloor$ edges. Consequently $k_t \leq nt/q$.

Given a feasible $k > 0$, set $t = \lfloor qk/n \rfloor$. For all sufficiently large n this is a valid choice, with $t \leq n/2$, and $k_t \leq k$. Add any $k - k_t$ missing edges and give each new edge weight one. A newly created K_r contains a new edge, so the extended weights remain a cover. Both the excess and the cost increase by $k - k_t$. It follows from (3) that

$$\phi_r^*(n,k) \leq k - \frac{(r-1)(r-2)}{2r}\frac{k^2}{n^2} + O_r(k/n+1). \tag{4}$$

Proof of Theorem 1.3. If $k = o(n^2)$, (1) gives $\phi_r(n,k) \geq (1-o(1))k$. Starting with $T_q(n)$ and adding k edges gives a fractional cover of cost k, so $\phi_r(n,k) \leq \phi_r^*(n,k) \leq k$. This proves sufficiency for both functions.

If k is not $o(n^2)$, pass to a subsequence on which $k \geq \varepsilon n^2$ for some $\varepsilon > 0$. Dividing (4) by k shows that the upper limit along this subsequence is at most $1 - (r-1)(r-2)\varepsilon/(2r) < 1$. Neither ratio can therefore converge to one. □

For the optimal second-order coefficient, we unbalance the external parts. Set

$$y = \frac{2}{rq}, \qquad x_i = \frac{1}{(r-i)(r-i+1)} + \frac{1}{rq} \quad (1 \leq i \leq q), \qquad Q_r = \sum_{i=1}^{q} x_i^2. \tag{5}$$

The weights are increasing and sum to one. More precisely,

$$\sum_{i=1}^{\ell} x_i = \frac{1}{r-\ell} - \frac{r-\ell-1}{rq} \qquad (0 \leq \ell \leq q). \tag{6}$$

Take a clique S of size $t = o(n)$ and independent external parts of sizes

$$b_i = \frac{n}{q} - x_i t + \delta_i, \qquad \sum_i \delta_i = 0, \quad |\delta_i| \le 1,$$

where the b_i are integers summing to $n - t$. Join all distinct external parts and join S to every external vertex. Give the edges of S weight y, the edges from S to part i weight x_i, and all other edges weight zero. By (6), the least weight of a clique having j vertices in S is

$$\binom{j}{2} y + j \sum_{i=1}^{r-j} x_i = 1.$$

These weights form a fractional cover.

If k'_t is the edge excess of this graph and W_t is the cover cost, the choice of the part sizes gives the exact identity

$$k'_t - W_t = \frac{Q_r - y}{2} t^2 - \frac{1-y}{2} t + g_q(n) - \frac{1}{2} \sum_i \delta_i^2. \tag{7}$$

Indeed, $e(G) = \binom{n}{2} - \sum_i \binom{b_i}{2}$ and $W_t = y\binom{t}{2} + t \sum_i b_i x_i$; their linear rounding terms cancel. Using $1/(j^2(j+1)^2) = 1/j^2 + 1/(j+1)^2 - 2/(j(j+1))$, we obtain

$$Q_r = 2H_r^{(2)} - 3 + \frac{2}{r} + \frac{1}{rq}, \qquad \frac{q^2(Q_r - y)}{2} = A_r.$$

For $n \ll k \ll n^2$, again take $t = \lfloor qk/n \rfloor$. Balancing the external parts can only increase the edge count, so $k'_t \le k_t \le k$. Add $k - k'_t$ edges of weight one. Equation (7) retains its value, and $t = (q + o(1))k/n \to \infty$. We have proved

$$\phi_r(n,k) \le \phi_r^*(n,k) \le k - (A_r - o(1)) \frac{k^2}{n^2}. \tag{8}$$

The only rounding loss in the deficit is $O_r(t+1)$, which is $o(k^2/n^2)$ throughout the stated range.

Remark 2.4. There are finite triangle examples attaining their cover cost exactly. Let s be a positive even integer and $b \ge 3s$. For $G = K_{3s} \vee K_{b,b+s-1}$,

$$\nu_3(G) = \nu_3^*(G) = 3sb + \frac{5s^2 - 3s}{2}, \qquad e(G) - t_2(v(G)) - \nu_3(G) = s(s-1).$$

To verify the packing, split the larger external part into sets of sizes b and $s-1$. Use $3s$ distinct perfect matchings of the resulting $K_{b,b}$, assigning one to each clique vertex. Split the clique into three sets of size s. For each of the remaining $s-1$ external vertices, use one perfect matching in each of these three sets. Their complete graphs have one-factorizations because s is even. The unused clique edges form $K_{s,s,s}$, which decomposes into s^2 triangles by addition in $\mathbb{Z}_s$. The total number is the displayed value. The cover with weights $1/3$ on clique edges, $2/3$ toward the smaller external part, and $1/3$ toward the larger external part has the same cost.

3 Fractional packings

The fractional proof reduces an optimal cover to a weighted complete graph with a small exceptional set. We first optimize this model, then control the number of exceptional vertices which contribute to the deficit.

The next lemma bounds the deficit in terms of the exceptional-set size and retains a quadratic penalty for imbalance of the external parts. The latter term is needed to control the cost of the later reductions. The weights may vary between exceptional vertices. We write $C_r = A_r/q^2 = H_r^{(2)} - 3/2 + 1/r - 1/(2rq)$.

Lemma 3.1. *Partition an n-vertex set into $S, A_1, \dots, A_q$, where $s = |S| \ge 1$ and $b_i = |A_i| > 0$. Suppose weights $w_{uv} \in [0,1]$ on all vertex pairs cover every r-set, are one inside each A_i, and are zero between distinct A_i. For $v \in S$, suppose $w_{vu} = a_i(v)$ for every $u \in A_i$, where $\sum_i a_i(v) = 1$. Define*

$$\mu_i = \frac{1}{s}\sum_{v\in S} a_i(v), \qquad D_{\mathrm{mod}} = \sum_{uv}(1 - w_{uv}) - \rho n^2.$$

Then

$$D_{\mathrm{mod}} \le C_r s^2 - \frac{1}{2}\sum_{i=1}^{q}(b_i - n/q + s\mu_i)^2. \tag{9}$$

Proof. Let $F_S = \sum_{uv\subset S}(1 - w_{uv})$ and $f = F_S/s^2$. For a set I of $r - j$ external indices and a j-set $U \subset S$, the covering condition reads

$$\sum_{uv\subset U} w_{uv} + \sum_{v\in U}\sum_{i\in I} a_i(v) \ge 1.$$

If $s \ge j$, average over all such U. With $\mu_I = \sum_{i\in I}\mu_i$, this gives

$$f \le \frac{s-1}{s}\left(\frac{1}{2} - \frac{1}{j(j-1)} + \frac{\mu_I}{j-1}\right) \le \frac{1}{2} - \frac{1}{j(j-1)} + \frac{\mu_I}{j-1}.$$

The second step uses the nonnegativity of the parenthesized expression. If $s < j$, the same final bound follows from $f \le (s-1)/(2s)$ and $j(j-1) \ge 2s$. Thus, for every $2 \le j \le r$,

$$f \le \frac{1}{2} - \frac{1}{j(j-1)} + \frac{\mu_I}{j-1}. \tag{10}$$

Relabel the parts so that $\mu_1 \le \cdots \le \mu_q$, and set $z = 1/2 - f$. Taking $j = r$ shows $1/(rq) \le z \le 1/2$. For $1 \le \ell < q$, equation (10), with $j = r - \ell$ and $I = \{1, \dots, \ell\}$, gives

$$\sum_{i=1}^{\ell}\mu_i \ge L_\ell(z) := \frac{1}{r-\ell} - (r - \ell - 1)z.$$

Set $S_0^* = 0$, $S_q^* = 1$, and $S_\ell^* = \max\{0, L_\ell(z)\}$ for $1 \le \ell < q$. The increments $\mu_i^* = S_i^* - S_{i-1}^*$ are nonnegative and increasing, since $L_\ell(z)$ is increasing and convex in ℓ. They sum to one. If $D_\ell = \sum_{i=1}^{\ell}(\mu_i - \mu_i^*)$, then $D_\ell \ge 0$ and $D_q = 0$. Summation by parts yields

$$\sum_i \mu_i^2 - \sum_i (\mu_i^*)^2 = \sum_{\ell=1}^{q-1} D_\ell(\mu_\ell + \mu_\ell^* - \mu_{\ell+1} - \mu_{\ell+1}^*) \le 0.$$

It remains to maximize $-z + \frac{1}{2}\sum_i(\mu_i^*)^2$. If $1/(J(J+1)) \le z \le 1/(J(J-1))$, where $2 \le J \le q$, its nonzero coordinates, allowing a zero at an endpoint, are $1/J - (J-1)z$ followed by $z + 1/(j(j-1))$ for $j = J, J-1, \dots, 2$. On this interval differentiation gives

$$\frac{d}{dz}\left(-z + \frac{1}{2}\sum_i(\mu_i^*)^2\right) = J(J-1)z - 1 \le 0.$$

These intervals cover $[1/(rq), 1/2]$, and the function is continuous at their endpoints. Its maximum therefore occurs at $z = 1/(rq)$. At this point $\mu_i^* = x_i$ from (5), so the maximum is $-1/(rq) + Q_r/2 = C_r$.

Finally, counting the weights between the parts and completing squares gives

$$\begin{aligned} D_{\mathrm{mod}} &= F_S - \frac{s^2}{2}\left(1 - \sum_i \mu_i^2\right) - \frac{1}{2}\sum_i (b_i - n/q + s\mu_i)^2 \\ &= s^2\left(-z + \frac{1}{2}\sum_i \mu_i^2\right) - \frac{1}{2}\sum_i (b_i - n/q + s\mu_i)^2. \end{aligned}$$

The preceding optimization proves (9). □

We now return to an arbitrary graph. We first locate a small exceptional set and normalize the cover to apply Lemma 3.1. To obtain the sharp coefficient, we then bound the number of exceptional vertices which contribute to the deficit.

Let $e(G) = t_q(n) + k$, and write $D = e(G) - \nu_r^*(G) - \rho n^2$. If $D \le 0$, then $\nu_r^*(G) \ge k - g_q(n)$, which is sufficient for Theorem 1.2. We therefore assume $D > 0$. Apply Lemma 2.1 to obtain G_0 and an optimal cover w with no edge of weight one. Put $k_0 = e(G_0) - t_q(n)$. Then $0 < k_0 \le k$, $\nu_r^*(G_0) < k_0$, and the value of D is unchanged.

Let F consist of the zero-weight edges of G_0. The graph F is K_r-free, and Lemma 2.1 gives

$$e(F) \ge t_q(n) - \left(\binom{r}{2} - 1\right) k_0.$$

For $k \le c_r n^2$, with $c_r > 0$ sufficiently small and n sufficiently large, Lemma 2.2 applies. It provides sets $A_1, \dots, A_q$ and a remaining set S such that

$$s = |S| = O_r((k+1)/n). \tag{11}$$

In fact, each A_i is independent in G_0. An edge of $G_0[A_i]$, together with the defining $(q-1)$-clique in F, would form a K_r all of whose other edges have weight zero. The edge would have weight one, contrary to its deletion.

If $s = 0$, the graph G_0 is q-partite and $D \le 0$, a contradiction. Thus $s \ge 1$. Put $b_i = |A_i|$. The inequality $e(G_0) \le n^2/2 - \frac{1}{2}\sum_i b_i^2$, together with $e(G_0) > \rho n^2$, implies $b_i = n/q + O_r(\sqrt{sn})$. By reducing c_r if necessary, we may assume $b_i \ge n/(2q)$ and $s/b_{\min} \le 1/4$, where $b_{\min} = \min_i b_i$.

To place the cover in the weighted model, we normalize its external weights while keeping track of the change in cost. Extend w to all pairs by assigning weight one to each nonedge of G_0. Every r-set now has total weight at least one, and $\sum_{uv}(1 - w_{uv}) = \rho n^2 + D$. The pairs from a core vertex to one external part form an arm. Define

$$L_{ij} = \sum_{u \in A_i,\ v \in A_j} w_{uv}, \qquad L = \sum_{i<j} L_{ij}, \qquad \ell_{ij} = \frac{L_{ij}}{b_i b_j},$$

and, for $v \in S$, define $\alpha_i(v) = b_i^{-1}\sum_{u \in A_i} w_{vu}$ and $\xi_i(v) = 1 - \alpha_i(v)$. Average over permutations within each external part. For a set $U \subset S$ and an external index set I with $|U| + |I| = r$, the resulting cover satisfies

$$\sum_{uv \subset U} w_{uv} + \sum_{v \in U}\sum_{i \in I} \alpha_i(v) + \sum_{\substack{i<j \\ i,j \in I}} \ell_{ij} \ge 1. \tag{12}$$

Set $c_i = \frac{1}{2}\sum_{j\neq i}\ell_{ij}$. Replace all external cross-weights by zero and replace $\alpha_i(v)$ by $\beta_i(v) = \min\{1, \alpha_i(v) + c_i\}$. This remains a cover. Indeed, an r-set with two vertices in the same external part contains a weight-one edge. Every other r-set has at least one vertex in S. If one of its arm weights reaches one it is covered; otherwise its added arm weight is at least $\sum_{i\in I} c_i \geq \sum_{i<j,\ i,j\in I}\ell_{ij}$. The cross-weight deletion saves L, while the added arm cost is at most $sL/b_{\min}$.

The one-core-vertex constraints give $\sum_i \beta_i(v) \geq 1$. Let $\delta_v = \sum_i \beta_i(v) - 1$, and reduce the arm weights to $a_i(v) \leq \beta_i(v)$ with $\sum_i a_i(v) = 1$. Increase the weight of each pair $uv \subset S$ by $\delta_u + \delta_v$, truncating at one. For an r-set with at least two core vertices, either a core edge reaches one or the increase in its core weights compensates for all decreased arm weights. Sets with one core vertex have arm sum one. The normalized object is therefore feasible. If its deficit is D_0, then

$$D_0 - D \geq (1 - s/b_{\min})L + (b_{\min} - s + 1)\sum_{v\in S}\delta_v.$$

Here arm reduction saves at least $b_{\min}\sum_v \delta_v$, and the core-weight increase costs at most $(s-1)\sum_v \delta_v$.

Apply Lemma 3.1 to this normalized object. Since $D_0 \geq D > 0$, it follows that

$$L = O_r(s^2), \qquad \sum_v \delta_v = O_r(s^2/n), \qquad b_i = n/q + O_r(s). \tag{13}$$

We keep track of the original arm quantities as well. The inequalities $\alpha_i \leq \beta_i$ and $\sum_i(\beta_i - a_i) = \delta_v$ imply

$$\xi_i(v) \geq 1 - a_i(v) - \delta_v, \qquad \sum_i \xi_i(v) \geq q - 1 - \delta_v. \tag{14}$$

On the other hand, (12) for one core vertex gives

$$\sum_i \xi_i(v) \leq q - 1 + \sum_{i<j}\ell_{ij} = q - 1 + O_r(L/n^2). \tag{15}$$

These relations concern the original graph, not only the normalized model.

The vertices whose arm vectors are close to a standard unit vector can be absorbed into an external part. We first bound the number of remaining vertices, using their neighborhoods in the original graph. Set $\eta = \sqrt{s/n}$, and choose c_r small enough that $\eta < 1/2$. Let $I = \{v \in S : \max_i a_i(v) \leq 1-\eta\}$ and $u = |I|$.

Claim 3.2. *With the notation above,*

$$u \leq \frac{qk_0}{n} + O_r(s\eta). \tag{16}$$

Proof. We bound the total number of missing external neighbors of vertices in I. For each $v \in I$, write $p_i = |N_{G_0}(v) \cap A_i|$ and $H_v = \sum_i(b_i - p_i)$.

There are $O_r(s^2/(n\eta))$ vertices with $\delta_v > \eta/2$. For every other $v \in I$, (14) gives $\xi_i(v) \geq \eta/2$. In the original extended cover, nonedges have weight one and hence $\sum_{x\in N_{G_0}(v)\cap A_i} w_{vx} = p_i - b_i\xi_i(v)$. In particular, $p_i \geq b_i\xi_i(v) \geq cn\eta$, where $c > 0$ depends only on r. Averaging its covering constraint over a choice of one actual neighbor in each part gives

$$\sum_i \frac{b_i\xi_i(v)}{p_i} \leq q - 1 + \sum_{i<j}\frac{L_{ij}}{p_ip_j}.$$

Subtracting the lower bound in (14), we obtain

$$\sum_i \xi_i(v)\left(\frac{b_i}{p_i} - 1\right) \leq \delta_v + \sum_{i<j}\frac{L_{ij}}{p_ip_j}.$$

The number of missing external neighbors therefore satisfies

$$H_v = O_r\left(\frac{n\delta_v}{\eta} + \frac{L}{n\eta^3}\right).$$

Summing over the vertices with $\delta_v \le \eta/2$ and bounding H_v by n for the remaining vertices yields

$$\sum_{v\in I} H_v = O_r\left(\frac{s^2}{\eta} + \frac{s^3}{n\eta^3}\right) = O_r(sn\eta).$$

Put $X_v = \sum_i b_i\xi_i(v)$. Equations (13)–(15) imply

$$\sum_{v\in S} X_v \ge \frac{q-1}{q}ns - O_r(s^2), \qquad X_v \le \frac{q-1}{q}n + O_r(s).$$

Every core vertex has external degree at least X_v. For $v \in I$, its external degree $n - s - H_v$ improves this bound by at least $n/q - O_r(s) - H_v$. The number of external edges is at least $\sum_{i<j} b_ib_j - L = \rho(n-s)^2 - O_r(s^2)$. Adding these estimates and discarding the nonnegative number of core edges, we obtain

$$e(G_0) \ge \rho n^2 + \frac{un}{q} - O_r(s^2 + sn\eta).$$

Since $e(G_0) = \rho n^2 + k_0 - g_q(n)$ and $s^2 \le sn\eta$, this proves (16). □

It remains to absorb the vertices outside I at a sufficiently small cost, so that the bound in Claim 3.2 can replace the original core size in Lemma 3.1. For each $v \in S \setminus I$, let i be the unique index with $a_i(v) > 1 - \eta$, and replace its arm vector by the standard unit vector e_i. The total arm weight decreased at v is $\gamma_v = 1 - a_i(v) < \eta$; the same amount is added to coordinate i. Because $b_i = n/q + O_r(s)$, the total change in arm cost over all these vertices has absolute value $O_r(\eta s^2)$. Set $\gamma_v = 0$ for $v \in I$, and increase every core-pair weight by $\gamma_u + \gamma_v$, truncating at one. The argument used for normalization shows that this repairs all covering constraints, at additional cost $O_r(\eta s^2)$.

A vertex with arm vector e_i can now be absorbed into A_i. To justify this, its edge to a vertex $z \in I$ has weight at least $a_i(z)$, as follows by completing the pair with one vertex from every external part other than A_i. Two vertices of type i have mutual weight one by the same argument. We may therefore reduce the weight from a type-i vertex to $z \in I$ to $a_i(z)$, and reduce all weights between different types to zero. Feasibility is preserved. An r-set with two vertices in the same enlarged external part contains a weight-one edge. In any other r-set, each external part occurs at most once, so we may replace every absorbed vertex by a representative of its original external part without identifying vertices. The resulting weights are those of a previously covered r-set.

The final normalized model has core I. If $u > 0$, Lemma 3.1 bounds its deficit by C_ru^2. If $u = 0$, its deficit is at most zero by the weighted edge count of a complete q-partite graph. In either case,

$$D \le C_ru^2 + O_r(\eta s^2). \tag{17}$$

Proof of Theorem 1.2. Choose c_r as required in the preceding reductions. For $k \ge n$, equations (11) and (16) give, with $h = k/n$,

$$s = O_r(h), \qquad u \le qh + O_r(h\sqrt{h/n}).$$

Substitution in (17) yields

$$D \le A_rh^2 + O_r(h^2\sqrt{h/n}) = A_r\frac{k^2}{n^2} + O_r\left(\frac{k^{5/2}}{n^3}\right).$$

For $k < n$, the same initial reduction has $s = O_r(1)$. The inequality $D \le D_0 \le C_r s^2$ directly gives $D = O_r(1)$. Finally, $k - \nu_r^*(G) = D + g_q(n)$ and $g_q(n) = O_r(1)$. Enlarging B_r also covers the finitely many smaller values of n. This proves the uniform statement. □

For $n \ll k \ll n^2$, the error terms in Theorem 1.2 are $o(k^2/n^2)$. Together with (8), this proves the fractional assertion of Theorem 1.1.

4 Integer packings

We now assume $n \ll k \ll n^2$ and put $h = k/n$. All statements in this section are for sufficiently large n along the chosen sequence. The integer proof requires a different initial reduction, since deleting weight-one edges of a fractional cover need not preserve the integer deficit. Instead we absorb all low-degree internal edges into an arbitrary packing of the graph left after their deletion.

Let G have $t_q(n) + k$ edges. We may assume $\nu_r(G) < k$. Delete the edges of a maximum packing to obtain a K_r-free graph F. Then $e(F) \ge t_q(n) - (\binom{r}{2} - 1)k$. Lemma 2.2 provides independent sets $A_1, \dots, A_q$ in F and an exceptional set S_* of size $O_r(h)$. There are $O_r(k)$ internal edges of G in these parts, because each was deleted from the packing. There are also $O_r(k)$ missing cross-edges. To see this, discard the $O_r(|S_*|n) = O_r(k)$ edges incident to S_* from F and compare the remaining edge count with $\rho(n - |S_*|)^2$. The same comparison shows $|A_i| = n/q + o(n)$ for every i.

Fix $\eta_0 = 1/(100q^3)$. Move to the core every external vertex whose internal degree plus missing cross-degree is at least $\eta_0 n$. The sum of these degrees is $O_r(k)$, so only $O_r(h)$ vertices are moved. Write S for the resulting core, $s = |S| = O_r(h)$, and $B_1, \dots, B_q$ for the remaining parts, of sizes $b_i \sim n/q$. Every remaining vertex has internal degree and missing cross-degree less than $\eta_0 n$. Let L be the set of all internal edges of the B_i, put $\ell = |L|$, and set $G_0 = G - L$.

Lemma 4.1. *For the partition above, $\nu_r(G) \ge \ell + \nu_r(G_0)$.*

Proof. Fix a packing $\mathcal{P}$ in G_0. Every member contains a vertex of S, since $G_0 - S$ is q-partite. Consequently every external vertex belongs to at most s members of $\mathcal{P}$, each using a different edge from that vertex to S. The used cross-edge degree is therefore at most $(q-1)s$.

Process the edges of L one at a time. For an edge uv in part B_j, choose one auxiliary vertex in each other part, forming a K_r. Require every vertex to be used as an auxiliary vertex at most $\lceil \eta_0 n \rceil$ times. Its number of appearances as an endpoint of an internal edge is less than $\eta_0 n$, so previously added cliques use cross-edges of maximum degree at most $2q\eta_0 n + q$.

When choosing an auxiliary vertex in a specified part, nonedges or used edges to the at most $q+1$ vertices already chosen exclude at most

$$(q+1)\big((2q+1)\eta_0 n + (q-1)s + q\big)$$

candidates. The total number of vertices already at the auxiliary-load limit is at most $(q-1)\ell/\lceil \eta_0 n \rceil = O_r(h) = o(n)$. Our choice of η_0, together with $b_i \sim n/q$ and $s = o(n)$, makes the total smaller than b_i. The selection can therefore always be completed. Each added clique contains exactly the internal edge currently processed, so it cannot use another unprocessed edge of L. This constructs ℓ new cliques disjoint in edges from one another and from $\mathcal{P}$. □

Since $\nu_r(G) < k$, Lemma 4.1 implies $k_0 := k - \ell > 0$ and $\nu_r(G_0) < k_0$. In particular $s \ge 1$, since otherwise G_0 would be q-partite with more than $t_q(n)$ edges.

The internal edges have now been absorbed. To control the missing external edges, we use an elementary matching estimate together with the small integer packing number.

Lemma 4.2. *Start with a complete q-partite graph whose part sizes are $p_1, \dots, p_q$. Delete at most M edges and then a graph of maximum degree at most D. The resulting graph has at least $\min_i p_i - O_q(D + \sqrt{M} + 1)$ vertex-disjoint copies of K_q.*

Proof. If $M > 0$, delete the at most $2\sqrt{M}$ vertices incident to more than $\sqrt{M}$ edges from the first deletion. The remaining missing-edge graph has maximum degree at most $D + \sqrt{M}$. Trim every part to the same size $m = \max\{0, \min_i p_i - \lceil 2\sqrt{M} \rceil\}$. If $m \geq 2(q-1)(D + \sqrt{M})$, join the parts one by one. At step j, the auxiliary bipartite graph between the already formed $(j-1)$-cliques and the next part has minimum degree at least $m - (j-1)(D + \sqrt{M}) \geq m/2$. Hall's theorem gives a perfect matching, and hence a K_q-factor. If the size inequality fails, the empty packing already meets the claimed bound. When $M = 0$, no initial vertices are deleted and the same argument applies. □

Let M be the number of missing edges between the parts B_i in G_0. For $v \in S$, set $p_{vi} = |N_{G_0}(v) \cap B_i|$ and $m_v = \min_i p_{vi}$, and put $B = \sum_{v \in S} m_v$. Choose $j(v)$ with $p_{v,j(v)} = m_v$. Write $t_j = |\{v \in S : j(v) = j\}|$, let $H = G_0[S]$, and define $T = \sum_{v \in S} \sum_{i \neq j(v)} (b_i - p_{vi})$. The next estimate controls both the missing edges and the imbalance of the external parts.

Claim 4.3. *With the notation above,*

$$M = O_r(s^2), \qquad T = O_r(s^2), \qquad b_i = n/q + O_r(s). \tag{18}$$

Proof. Apply Lemma 4.2 successively to the external neighborhoods of the core vertices. Each previous core vertex contributes a vertex-disjoint family of transversal cliques, so the used external-edge graph has maximum degree at most $(q-1)s$. Extending the transversal cliques by their core vertex gives

$$\nu_r(G_0) \geq B - O_r(s^2 + s\sqrt{M}). \tag{19}$$

The additive $O_r(s)$ from the lemma is absorbed by s^2, since $s \geq 1$.

Expanding the edge count and completing squares gives

$$k_0 - B = e(H) - M - T - \frac{s^2 - \sum_j t_j^2}{2} - \frac{1}{2} \sum_i (b_i + t_i - n/q)^2 + g_q(n). \tag{20}$$

For completeness, the external edge count is $((n-s)^2 - \sum_i b_i^2)/2 - M$, and the core-to-external edge count minus B is $s(n-s) - \sum_j t_j b_j - T$. Adding $e(H)$ and subtracting $t_q(n) = \rho n^2 - g_q(n)$ yields (20).

Combine $\nu_r(G_0) < k_0$ with (19) and (20). Since $e(H) \leq s(s-1)/2$, $s^2 - \sum_j t_j^2 \geq 0$, and $g_q(n) = O_r(1)$, there is a constant C depending only on r such that

$$M + T + \frac{1}{2} \sum_i (b_i + t_i - n/q)^2 \leq C(s^2 + s\sqrt{M}).$$

Solving the quadratic inequality in $\sqrt{M}/s$ first, and then returning to this estimate, proves (18). □

Claim 4.3 allows us to complete the external graph and all but one neighborhood per core vertex by adding only $O_r(s^2)$ edges. The completed graph has the structure needed for compression. Complete all edges between distinct external parts. For each $v \in S$, complete its edges to every part except $B_{j(v)}$. This adds $M + T = O_r(s^2)$ edges. If some $i \neq j(v)$ satisfies $b_i - m_v < 2q^2 s$, complete the remaining part as well. This costs at most $|b_{j(v)} - b_i| + 2q^2 s = O_r(s)$ per vertex by (18).

Call the resulting graph G_1, and let $a = O_r(s^2)$ be the total number of added edges. Partition its core into $U, X_1, \dots, X_q$. The vertices of U are joined to every external vertex. Each $v \in X_j$ is joined to every part other than B_j, has m_v neighbors in B_j, and satisfies

$$m_v \le \min_{i \ne j} b_i - 2q^2 s. \tag{21}$$

Vertices already complete to all parts are included in U. Adding an edge increases the maximum integer packing by at most one, as at most one member of a packing can contain that edge. For $k_1 = k_0 + a$ we therefore have

$$k - \nu_r(G) \le k_1 - \nu_r(G_1), \qquad 0 < k_1 \le k + O_r(h^2) = k + o(k). \tag{22}$$

It remains to round the packing at the scale of the core. The following proposition controls the fractional–integer gap of G_1 by a graph with $O_r(s)$ vertices. The argument keeps the arbitrary neighborhoods in the partial parts; no regularity assumption on them is needed.

Proposition 4.4. *Let G_1 have the structure described above, with independent external parts B_i spanning a complete q-partite graph and core $S = U \cup X_1 \cup \cdots \cup X_q$, where $s = |S| \ge 1$. Assume* (21)*, and put $b = \min_i b_i$ and $\Delta_i = b_i - b$. If $b \ge (q+4)s$, there is a graph J on $(4q+1)s + \sum_i \Delta_i$ vertices such that*

$$\nu_r^*(G_1) = B_* + |U|(b - 4s) + \nu_r^*(J), \qquad \nu_r(G_1) \ge B_* + |U|(b - 4s) + \nu_r(J),$$

where $B_* = \sum_{v \in S \setminus U} m_v$. In particular, $\nu_r^*(G_1) - \nu_r(G_1) \le \nu_r^*(J) - \nu_r(J)$.

Proof. Set $R = 4s$ and $u = |U|$. The graph J retains $H = G_1[S]$. Its external part i has $R + \Delta_i$ vertices, and distinct external parts are completely joined. Every vertex of U is joined to all parts. A vertex in X_j is joined to all parts other than j and to no vertex in part j.

Fractional compression. For a fractional packing, aggregate each clique containing at least two core vertices by its core set T and its set I of external parts. Write $y_{T,I}$ for the total weight of that type. Thus $H[T]$ is complete, $2 \le |T| \le r$, and $|I| = r - |T|$. The core-edge constraints are

$$\sum_{\substack{(T,I) \\ e \subset T}} y_{T,I} \le 1 \qquad (e \in E(H)). \tag{23}$$

Define $l_{vi} = \sum_{(T,I):\, v \in T,\ i \in I} y_{T,I}$. Since each such clique uses a core edge at v,

$$l_{vi} \le \sum_{(T,I):\, v \in T} y_{T,I} \le s - 1, \qquad \sum_{T,I} y_{T,I} \le \binom{s}{2}. \tag{24}$$

For $v \in U$, the total weight of cliques whose only core vertex is v is at most $b - \pi_v$, where $\pi_v = \max_i(l_{vi} - \Delta_i)$. The number π_v lies in $[0, s-1]$, because $\min_i \Delta_i = 0$. For $v \in X_j$, that total weight is at most $m_v - l_{vj}$. It follows that every fractional packing has total weight at most

$$B_* + ub + \sum_{T,I} y_{T,I} - \sum_{v \in U} \pi_v - \sum_j \sum_{v \in X_j} l_{vj}.$$

If a type uses both a vertex of X_j and external part j, delete its weight from y. The positive term decreases by that weight, but the last sum decreases by at least the same amount, and no

π_v increases. Hence this deletion cannot decrease the displayed upper-bound objective. We may restrict to types satisfying

$$v \in T \cap X_j \implies j \notin I. \tag{25}$$

Let Q be the maximum of $\sum y_{T,I} - \sum_{v \in U} \pi_v$ over nonnegative weights satisfying (23) and (25). The feasible set is compact by (24), including when it has no types. We have proved $\nu_r^*(G_1) \le B_* + ub + Q$.

This upper bound is attainable. Take maximizing weights $y_{T,I}$ and distribute each uniformly over the product of the indicated external parts. All these parts are completely joined to every vertex of T by (25). For each $v \in U$, distribute single-core weight $b - \pi_v$ uniformly over all external transversals. For each $v \in X_j$ with $m_v > 0$, distribute single-core weight m_v uniformly over its actual neighborhood in B_j and all other parts. When $m_v = 0$, there is no such weight.

The core-edge loads satisfy (23). An arm edge at $v \in U$ in part i has load $(b - \pi_v + l_{vi})/b_i \le 1$. The partial arm edges at $v \in X_j$ have load one, and its complete arm edges have load $(m_v + l_{vi})/b_i \le 1$ by (21). For a fixed external edge, the contribution from any one core vertex's single-core cliques is at most $1/b$. Indeed, for a partial vertex and an edge incident to its partial part the contribution is either zero or $1/b_i$; for the other external edges it is at most $m_v/(b_i b_{i'}) \le 1/b$. The contribution from all multi-core cliques is at most $\binom{s}{2}/b^2$ by (24). Thus the external edge load is at most $s/b + \binom{s}{2}/b^2 < 1$. The packing is feasible and has weight $B_* + ub + Q$.

The same allowed types and core capacities govern J. For its full vertices the baseline is R, and for its partial vertices the baseline is zero. The preceding construction applies with b replaced by $R = 4s$: partial vertices have only multi-core load at most $s - 1$ on any complete arm, and the external load is at most $s/R + \binom{s}{2}/R^2 < 1$. Consequently $\nu_r^*(J) = uR + Q$, proving the asserted fractional identity.

Integer extension. Embed any packing of J into G_1 using $R + \Delta_i$ vertices of B_i. Each unused part then has the same size $m = b - R$. Label its vertices by $\mathbb{Z}_m$ and number the parts $0, \dots, q-1$. For each $\ell \in \{0, \dots, u-1\}$, the transversals

$$(x, x + \ell, \dots, x + (q-1)\ell), \qquad x \in \mathbb{Z}_m,$$

form a K_q-factor. Different factors have no common edge. A common edge between parts $i < j$ would imply $(j-i)(\ell - \ell') \equiv 0 \pmod m$, whereas $0 < |(j-i)(\ell - \ell')| \le (q-1)(u-1) < m$. Assigning one factor to each vertex of U therefore adds um cliques. No divisibility assumption on m is used.

Next process the vertices $v \in X_j$ one by one. All their m_v partial-arm edges are still unused, since J has no such edges and the factors just added use vertices of U. Every clique of the embedded packing containing v uses an edge from v to another core vertex. There are at most $s - 1$ such cliques in the packing, so together they use at most $s - 1$ edges of any complete arm at v. For each actual neighbor of v in B_j, select one vertex in each other part to complete a clique, taking distinct vertices within a given part for different neighbors of this same v.

At every stage, each external vertex belongs to at most s packed cliques: each clique uses a distinct edge from it to the core. Hence its used degree toward any one other external part is at most s. When selecting a vertex in a complete part B_i, the initially used arm edges exclude at most $s - 1$ candidates, previous selections for this v exclude at most $m_v - 1$, and used edges to the at most $q - 1$ already selected external vertices exclude at most $(q-1)s$. The number of available choices is therefore at least

$$b_i - (s-1) - (m_v - 1) - (q-1)s = b_i - m_v - qs + 2 > 0$$

by (21). The procedure adds all B_* required cliques. They use no edge of the embedded packing or of the full-vertex factors. Together with the embedded packing, the added cliques give $\nu_r(G_1) \ge \nu_r(J) + u(b - 4s) + B_*$, as required. □

Proof of the integer assertion of Theorem 1.1. For a graph G with $t_q(n)+k$ edges, the case $\nu_r(G) \geq k$ is immediate. Otherwise the preceding reductions give G_1 with $s = O_r(h)$, $\Delta_i = O_r(s)$, and (22). Since $s = o(n)$ and $b_i \sim n/q$, Proposition 4.4 applies for sufficiently large n. Its graph J has $O_r(h)$ vertices. Pad J with isolated vertices to $\lceil Ch \rceil$ vertices, for a fixed sufficiently large C depending only on r. This changes neither packing number, and $h \to \infty$. Theorem 2.3, used on this smaller graph, gives

$$0 \leq \nu_r^*(G_1) - \nu_r(G_1) \leq \nu_r^*(J) - \nu_r(J) = o_r(h^2).$$

The excess k_1 may be smaller than n, so we use the uniform statement of Theorem 1.2, rather than only its asymptotic corollary. Equation (22) gives

$$k - \nu_r(G) \leq k_1 - \nu_r(G_1) \leq A_r(k_1/n)^2 + B_r k_1^{5/2}/n^3 + B_r + o_r(h^2) \leq (A_r + o(1))h^2.$$

For the last inequality, $k_1 \leq k + O_r(h^2)$ implies $k_1/n \leq h + O_r(h^2/n)$, while $h/n \to 0$ and $h \to \infty$. This is the required lower bound for every graph. The construction (8) supplies the reverse inequality at every feasible edge count, proving the theorem. □

5 Concluding remarks

Theorem 1.1 determines the second-order term for every fixed clique size throughout $n \ll k \ll n^2$. The integer proof uses uniform Haxell–Rödl rounding on the compressed graph, so it does not give the explicit remainder in the fractional bound of Theorem 1.2. Determining the optimal smaller-order error for either packing problem remains open. At $k = \Theta(n)$ the deficit scale is bounded, and the cases outside the known exact ranges require a finer analysis. The fixed positive-density problem $k = \Theta(n^2)$ also remains open.

Proposition 4.4 isolates the rounding mechanism used here. Its hypotheses retain one arbitrary external neighborhood per core vertex, with a capacity margin on the other parts. Determining which broader structures admit such a compression, and describing all graphs attaining the second-order bound asymptotically, are natural further questions.

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